

Revision Games Pack — 1.4 Data Types, Data Structures & Boolean Algebra

OCR A Level Computer Science (H446) · Component 01 · 1.4.1 Data types · 1.4.2 Data structures · 1.4.3 Boolean algebra

A printable set of four games. Cut out cards where indicated. All numeric answers have been verified.

Loop cards (dominoes)

A closed loop of **15 cards** mixing definitions and quick conversions. Each card states a value/term ("I have...") and asks a question ("Who has..."). The answer to every question is the "I have" on another card; the chain returns to the start.

How to play 1. Cut out the 15 cards and deal them so every student has at least one (some may hold two). 2. Pick any starter — read its "Who has..." question aloud. 3. The student whose card answers it reads "I have... Who has...", and play continues. 4. The loop is complete when you return to the starting card. Time the class and try to beat it.

Card	I have...	Who has...
1	0xD6	...the denary value of binary 1010 ?
2	10	...the result of 0 + 0 in binary?
3	0	...the number of bits in a nibble?
4	4	...the denary value of hex F ?
5	15	...the denary value of binary 00100000 ?
6	32	...the hex value of binary 1111 ?
7	F	...the data structure that is LIFO?
8	Stack	...the data structure that is FIFO?
9	Queue	...the 8-bit two's complement of denary -1 ?
10	11111111	...the denary value of binary 00001000 ?
11	8	...the effect of a logical left shift by 1, as a multiplication?
12	Multiply by 2	...the law that states $\neg(A \cdot B) = \neg A + \neg B$?
13	De Morgan's	...the floating-point part that sets the precision?
14	Mantissa	...the value of 0xD6 (214) minus 200, in denary?
15	14	...the hex value of denary 214 ?

Loop closes: card 15's answer (214 in hex = D6 → 0xD6) leads back to card 1. ✓

Verification of every conversion: binary 1010 = 10 ✓ · 0+0 = 0 ✓ · nibble = 4 bits ✓ · hex F = 15 ✓ · binary 00100000 = 32 ✓ · binary 1111 = hex F ✓ · two's complement of -1 = 11111111 ✓ · binary 00001000 = 8 ✓ · left shift ×1 = ×2 ✓ · 214 - 200 = 14 ✓ · denary 214 = hex D6 ✓.

Taboo cards

How to play - One student describes the **TERM** at the top of the card to their team **without saying any of the 4 banned words** (or obvious word forms of them). - 30 seconds per card, 1 point per correct guess. If a banned word slips out, the card is forfeited. - Rotate the describer each round.

#	TERM	Banned words
1	Two's complement	negative · MSB · invert · add one
2	Normalisation	mantissa · precision · point · leading
3	Stack	LIFO · push · pop · top
4	De Morgan's	bar · NOT · swap · AND/OR
5	Hash table	key · function · collision · index
6	Linked list	node · pointer · next · traverse
7	Logical shift	bits · multiply · divide · zero
8	Binary search tree	left · right · subtree · ordered

Bingo — binary/hex number bingo

Students fill a **3×3 grid** by choosing any 9 values from the word bank below (written as the binary or hex **pattern**, not the denary). The caller reads a **denary value or a clue**; students cross off the matching pattern. First to a line (or full house) wins.

How to play 1. Hand out blank 3×3 grids. Each student writes 9 different items chosen from the 20-item word bank (copy them exactly, e.g. `binary 01100100` or `hex C8`). 2. The caller works down the numbered caller list, reading the **clue** (left column) aloud — NOT the answer. 3. Students cross off a cell only if its pattern equals the called value. 4. Call "line" for a row/column/diagonal, or "full house" for all nine. Verify against the answers column.

Word bank (20 items — write any 9 on your grid):

`binary 00001010` · `binary 00010100` · `binary 00011110` · `binary 00101000` · `binary 01100100` ·
`binary 11111111` · `binary 10000000` · `binary 01000000` · `binary 00010000` · `binary 00000001` ·
`hex 0F` · `hex 1E` · `hex 32` · `hex 64` · `hex FF` · `hex 80` · `hex 0A` · `hex 2D` · `hex C8` · `hex 19`

Caller list (read the clue; answer = the value students must match):

#	Caller reads (clue)	Value	Patterns that match
1	"Ten"	10	binary 00001010, hex 0A
2	"Twenty"	20	binary 00010100
3	"Thirty"	30	binary 00011110, hex 1E
4	"Forty"	40	binary 00101000
5	"One hundred"	100	binary 01100100, hex 64
6	"The largest value in one byte (unsigned)"	255	binary 11111111, hex FF
7	"Two to the power 7"	128	binary 10000000, hex 80
8	"Two to the power 6"	64	binary 01000000
9	"Two to the power 4"	16	binary 00010000
10	"The smallest positive whole number"	1	binary 00000001
11	"Hex F in denary"	15	hex 0F
12	"Fifty"	50	hex 32
13	"Forty-five"	45	hex 2D
14	"Two hundred"	200	hex C8
15	"Twenty-five"	25	hex 19

All values verified: e.g. $\text{hex } C8 = (12 \times 16) + 8 = 200 \checkmark$ · $\text{hex } 2D = (2 \times 16) + 13 = 45 \checkmark$ · $\text{hex } 19 = (1 \times 16) + 9 = 25 \checkmark$ · $\text{binary } 00011110 = 16 + 8 + 4 + 2 = 30 \checkmark$.

Quick-fire "Guess the term/value"

How to play - Read a clue; students write or shout the answer. Keep it fast — 10–15 seconds each. - Mix of definitions and one-step conversions; answers verified below.

#	Clue	Answer
1	The signed representation where the MSB has a negative place value.	Two's complement
2	Convert binary 11001010 to denary.	202
3	Convert denary 45 to hex.	2D
4	The float part that determines the range of representable values.	Exponent
5	A right shift that copies the sign bit in from the left.	Arithmetic (right) shift
6	LIFO data structure with push, pop and peek.	Stack
7	Convert hex BE to denary.	190
8	An ordered, immutable sequence such as (x, y) .	Tuple
9	The half-adder output given by $A \cdot B$.	Carry
10	Two keys mapping to the same index in a hash table.	Collision
11	Convert binary 1111 to hex.	F
12	An in-order traversal of a BST visits the nodes in this order.	Ascending order

Verification: binary 11001010 = $128+64+8+2 = 202$ ✓ · denary 45 = 0010 1101 = 2D ✓ · hex BE = $(11 \times 16) + 14 = 190$ ✓ · binary 1111 = hex F ✓ · half-adder carry = A·B ✓ · in-order of BST = ascending ✓.